

Mechanical integrity

feasible design  
window

low-pole industrial topology  
reachable with silicon IGBT

power-electronic interface: the gatekeeper

high-pole mobile topology  
feasible only with SiC or GaN

thicker sleeve  
→ larger air gap

cooling channels  
→ lower stiffness

more poles, higher frequency → more loss in less volume

Electromagnetic  
efficiency

Thermal  
management

